PR2 Worked Examples: Quadratic Models

Recording companion for the PR2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Is the vertex a maximum or a minimum, and does the question ask for its input or its output? The vertex answers both questions. Its input, $x = -b/(2a)$, says when or how many. Its output, found by putting that input into the model, says how high or how low. Then read the answer back in the story: add the start year to x, round units to whole numbers, and give the units.

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Example 1: A Projectile

$$h(t)=-16t^2+128t+144$$

Question

A projectile is fired into the air from an initial height of 144 feet. Its height after t seconds is h(t) feet. When does it reach its maximum height?

A. 8 seconds
B. −8 seconds
C. 4 seconds
D. 9 seconds

Answer and reasoning

The time of the maximum is the input of the vertex: $t=-b/(2a)=-128/(2(-16))=128/32=4$ seconds. 8 seconds is $-b/a$, which leaves out the 2; at $t=8$ the projectile is back down at 144 feet. −8 seconds is $b/a=128/(-16)$, which drops both the minus and the 2; and 9 seconds is when it lands, $h(9)=0$, not when it is highest. **4 seconds.**

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Example 2: Average Age

$$a(x)=0.0035x^2-0.42x+45.1$$

Question

In a town, the average age of adult residents x years after 1900 was a(x) years. In what year was the average age at a minimum? (Round to the nearest year.)

A. 1960
B. 60
C. 1780
D. 2020

Answer and reasoning

$a=0.0035$ is positive, so the parabola opens up and the vertex is a minimum. Its input: $x=-(-0.42)/(2(0.0035))=0.42/0.007=60$. But x counts years after 1900, so the year is $1900+60=1960$. 60 is the right x but not a year; 1780 makes two slips, dividing by a and dropping the minus, so $x=-120$; and 2020 divides by a instead of $2a$, giving $x=120$. **1960.**

Example 3: Unit Cost

$$C(x)=0.4x^2-152x+18000$$

Question

The cost per unit to produce x widgets is C(x) dollars. How many widgets should be produced to minimize the unit cost? (Round to the nearest unit.)

A. 380 widgets
B. 190 widgets
C. 3,800 widgets
D. 3,560 widgets

Answer and reasoning

$x=-(-152)/(2(0.4))=152/0.8=190$. With $a=0.4$ positive, that is a minimum. 380 is $152/0.4$, which leaves out the 2; 3,800 is $152/0.04$, which leaves out the 2 and also slips a decimal place; and 3,560 is $C(190)$, the least cost in dollars, which answers the other question. **190 widgets.**

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Example 4: A Half Second

$$h(t) = -16t^2 + 80t + 384$$

Question

A projectile is fired into the air from an initial height of 384 feet. Its height after t seconds is h(t) feet. When does it reach its maximum height?

A. 2.5 seconds
B. 5.0 seconds
C. 8.0 seconds
D. −5.0 seconds

Answer and reasoning

$t = -80/(2(-16)) = 80/32 = 2.5$ seconds. Half seconds are common on CheckIt, and 2.5 is already rounded to a tenth. 5 seconds is $80/16$, which leaves out the 2; 8 seconds is when it lands, $h(8) = 0$; and −5 seconds is $80/(-16)$, which drops both the minus and the 2. **2.5 seconds.**

Example 5: From the Ground

$$h(t)=-16t^2+160t$$

Question

A projectile is fired into the air from an initial height of 0 feet. Its height after t seconds is h(t) feet. What is the maximum height it reaches?

A. 800 feet
B. 400 feet
C. 720 feet
D. 0 feet

Answer and reasoning

First the time: $t=-160/(2(-16))=5$ seconds. Then the height: $h(5)=-16(25)+160(5)=-400+800=400$ feet. 800 feet is only the $160t$ term, leaving out $-16t^2$; 720 feet forgets to square the 5, using $-16(5)=-80$; and 0 feet is the height at launch, and again when it lands at $t=10$. **400 feet.**

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Example 6: Least Average Age

$$a(x)=0.0042x^2-0.34x+41.25$$

Question

In a town, the average age of adult residents x years after 1900 was a(x) years. What was the minimum average age? (Round to the nearest tenth of a year.)

A. 61.9 years
B. 27.7 years
C. 41.3 years
D. 34.4 years

Answer and reasoning

$x=0.34/(2(0.0042))=0.34/0.0084\approx 40.476$. Time can be any number, so keep x unrounded: $a(40.476)\approx 0.0042(1638.3)-0.34(40.476)+41.25\approx 6.881-13.762+41.25=34.369$, about 34.4 years. (CheckIt evaluates at the rounded year, $a(40)=34.37$, which also rounds to 34.4.) 61.9 adds the 13.762 instead of subtracting it; 27.7 forgets to square, using $0.0042(40.476)\approx 0.170$; and 41.3 is $a(0)=41.25$, the average age in 1900, not the least one. **34.4 years.**

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Example 7: Least Unit Cost

$$C(x)=0.3x^2-145x+20000$$

Question

The cost per unit to produce x widgets is C(x) dollars, and the number of widgets that minimizes it is rounded to the nearest unit. What is the minimum unit cost? (Round to the nearest cent.)

A. $2,479.20
B. $2,479.17
C. $20,000.00
D. $72,659.20

Answer and reasoning

$x=145/(2(0.3))=145/0.6\approx 241.67$, so make 242 widgets: you cannot make part of one. Then $C(242)=0.3(58{,}564)-145(242)+20{,}000=17{,}569.20-35{,}090+20{,}000=2{,}479.20$. CheckIt does the same: it evaluates the cost at the rounded number of units, and prints the answer as 2479.2. $2,479.17 is the cost at the exact vertex, 241.67 widgets, which is not a whole number of widgets; $20,000.00 is $C(0)$, the constant term; and $72,659.20 adds the 35,090 instead of subtracting it. **$2,479.20.**

Example 8: Greatest Height

$$h(t)=-16t^2+48t+288$$

Question

A projectile is fired into the air from an initial height of 288 feet. Its height after t seconds is h(t) feet. What is the maximum height it reaches?

A. 360 feet
B. 324 feet
C. 336 feet
D. 288 feet

Answer and reasoning

$t=48/32=1.5$ seconds. Then $h(1.5)=-16(2.25)+48(1.5)+288=-36+72+288=324$ feet. 360 feet leaves out the $-16t^2$ term; 336 feet forgets to square the 1.5, using $-16(1.5)=-24$; and 288 feet is the launch height, $h(0)$. **324 feet.**

Example 9: Why a Least Cost

$$C(x)=0.2x^2-101x+15000$$

Question

The cost per unit to produce x widgets is C(x) dollars. Why does this model have a minimum unit cost, and no maximum?

A. a = 0.2 is positive, so the parabola opens up
B. b = −101 is negative, so the cost goes down
C. c = 15,000 is the largest the cost can be
D. Costs cannot be negative, so it has a floor

Answer and reasoning

The sign of a, the number in front of x^2, decides the shape. $a=0.2$ is positive, so the parabola opens up: the cost falls to the vertex and then rises forever, with a least value and no greatest. A negative b only moves the vertex to the right of 0; the cost does go down at first, but it turns around; $C(0)=15{,}000$ is the cost at 0 units, and for large x the cost goes far past it; and costs that cannot be negative would not explain the turning point. ***a* is positive, so the parabola opens up.**

Example 10: Reading the Vertex

$$h(t)=-16t^2+96t+112$$

Question

A projectile's height after t seconds is h(t) feet. The vertex of h is (3, 256). What does the vertex tell you?

A. After 256 seconds it reaches its greatest height, 3 feet
B. It hits the ground after 3 seconds, 256 feet away
C. After 3 seconds it reaches its greatest height, 256 feet
D. It is launched from 256 feet and lands after 3 seconds

Answer and reasoning

The vertex is $(t,h(t))$: input first, a time in seconds; output second, a height in feet. With $a=-16$, it is the top of the path: 3 seconds after launch, the projectile is 256 feet high. Check: $h(3)=-144+288+112=256$. Reading 256 as the time swaps the input and the output; the projectile lands when $h(t)=0$, which is at $t=7$, and the model gives a height, not a distance away; and the launch height is $h(0)=112$, not 256. **After 3 seconds it reaches its greatest height, 256 feet.**

Example 11: Rounding the Year

$$a(x)=0.0026x^2-0.3x+44$$

Question

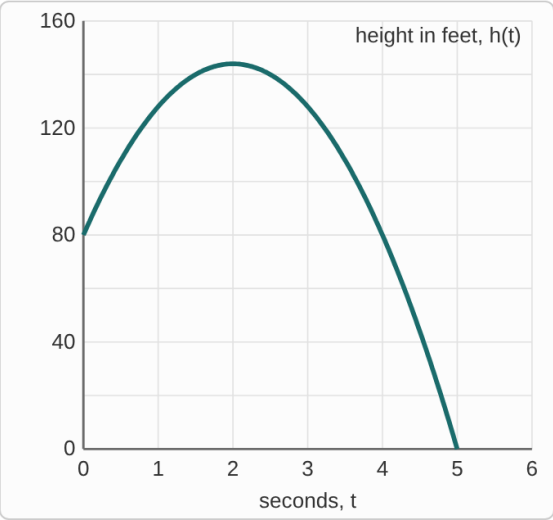
In a town, the average age of adult residents x years after 1900 was a(x) years. In what year was the average age at a minimum? (Round to the nearest year.)

A. 1957
B. 1958
C. 57
D. 1842

Answer and reasoning

$x=0.3/(2(0.0026))=0.3/0.0052\approx 57.69$. Rounded to the nearest whole year, $x=58$, so the year is $1900+58=1958$. 1957 cuts off the decimal instead of rounding, but 57.69 is closer to 58; 57 cuts off the decimal and also forgets to add 1900; and 1842 drops the minus in $-b/(2a)$, giving $x=-58$. **1958.**

Example 12: The Constant Term



Plotted data: A graph with seconds, t, from 0 to 6 across and height in feet, h(t), from 0 to 160 up. The curve starts at height 80 when t = 0, rises to its top point at t = 2, where the height is 144, and comes back down to 0 at t = 5.

$$h(t)=-16t^2+64t+80$$

Question

A projectile's height after t seconds is h(t) feet. What does h(0) = 80 tell you?

- **A.** Its greatest height is 80 feet
- **B.** It lands after 80 seconds
- **C.** It starts 80 feet above the ground
- **D.** It starts upward at 80 feet per second

Answer and reasoning

h(0) is the height at *t=0*, the moment of launch, so the projectile starts 80 feet up. CheckIt's stories say this as "an initial height of 80 feet." Its greatest height is at *t=2*: *h(2)=−64+128+80=144* feet, higher than the launch; it lands when *h(t)=0*, at *t=5*; and the upward speed at launch is the 64 in *64t*, feet per second, not the 80. **It starts 80 feet above the ground.**

Example 13: Slip 1

$$a(x)=0.003x^2-0.33x+46$$
$$x=0.33/(2(0.003))=55$$

Question

In a town, the average age of adult residents x years after 1900 was a(x) years. A student found the year of the minimum average age and answered "55". What is the mistake, if any?

A. There is no mistake.
B. Add the 1900: the year is 1955.
C. Divide by a, not 2a: the year is 2010.
D. The year is a(55), about 36.9.

Answer and reasoning

The vertex work is right: $x=55$. But the question asks for a year, and x counts years after 1900, so the year is $1900+55=1955$. The formula needs $2a$, and $0.33/0.003=110$ would give 2010; and $a(55)\approx 36.9$ is the least average age, which answers the other question. **1955.**

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Example 14: Slip 2

$$h(t)=-16t^2+112t+480$$
$$t=112/16=7$$
$$h(7)=480$$

Question

A student found the maximum height of this projectile and answered 480 feet. What is the mistake, if any?

A. There is no mistake.
B. The time is right, but h(7) is 1,264 feet.
C. t = −b/(2a) = 3.5, so the height is 676 feet.
D. The greatest height is the launch height, 480 feet.

Answer and reasoning

The formula is $t=-b/(2a)=-112/(2(-16))=3.5$ seconds; the student left out the 2. Then $h(3.5)=-16(12.25)+112(3.5)+480=-196+392+480=676$ feet. A clue: the student's answer equals the launch height, because $t=7$ is the mirror image of $t=0$ across the vertex. $h(7)=-784+784+480=480$, so 1,264, which is $112(7)+480$, leaves out the $-16t^2$ term; and the projectile rises above its launch height, so 480 is not the greatest. **676 feet, at 3.5 seconds.**

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Example 15: Slip 3

$$C(x)=0.6x^2-180x+25000$$
$$x=180/(2(0.6))=150$$
$$C(150)=13500-27000+25000=11500$$

Question

The cost per unit to produce x widgets is C(x) dollars. A student found the number of widgets that minimizes the unit cost, and the minimum unit cost. What is the mistake, if any?

A. There is no mistake.
B. a = 0.6 is positive, so 150 gives the greatest cost.
C. 2(0.6) is 0.12, so x should be 1,500.
D. The least cost is the constant, $25,000.

Answer and reasoning

$x=180/1.2=150$ widgets, a whole number already. $C(150)=0.6(22{,}500)-180(150)+25{,}000=13{,}500-27{,}000+25{,}000=11{,}500$. The student is right: produce 150 widgets, for a least unit cost of $11,500.00. A positive *a* opens up, so the vertex is the least cost, not the greatest; $2(0.6)=1.2$, and 0.12 is a decimal slip; and $25,000 is $C(0)$, which is more than $C(150)$. **There is no mistake.**

Example 16: Slip 4

$$a(x)=0.0045x^2-0.47x+48.2$$
$$x\approx 52.22$$
$$a(52.22)\approx 0.0045(104.44)-24.54+48.2\approx 24.1$$

Question

In a town, the average age of adult residents x years after 1900 was a(x) years. A student found the minimum average age. What is the mistake, if any?

A. There is no mistake.
B. 52.22 is past 52, so use a(53) for the least age.
C. The minimum is at x = 1952, so use a(1952).
D. Square 52.22 instead of doubling it: 35.9.

Answer and reasoning

$52.22^2\approx 2{,}727$; the student doubled 52.22 instead of squaring it. So $a(52.22)\approx 0.0045(2{,}727)-24.54+48.2\approx 12.27-24.54+48.2\approx 35.9$ years. The vertex is the least point, so the input stays at 52.22, not the next whole year, 53; and x is years after 1900, so the year 1952 is $x=52$, not $x=1952$. **35.9 years.**

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Example 17: Slip 5

$$h(t)=-16t^2+96t$$
$$vertex=(3,144)$$

Question

A projectile is fired from the ground. A student found the vertex and wrote, "The lowest the projectile gets is 144 feet, at 3 seconds." What is the mistake, if any?

A. There is no mistake.
B. 144 feet is the greatest height, not the least.
C. The vertex should be the point (−3, −432).
D. The vertex should be (6, 0), the point where it lands.

Answer and reasoning

The vertex is right: $t=96/32=3$ and $h(3)=-144+288=144$. But $a=-16$ is negative, so the parabola opens down and the vertex is the top: 144 feet is the greatest height. The projectile is lower than 144 feet at launch, $h(0)=0$. $(-3,-432)$ drops the minus in $-b/(2a)$; and $(6,0)$ is where it lands, a point on the parabola but not the vertex. **144 feet is the greatest height, at 3 seconds.**

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